

TIMING CHARACTERISTICS

(Unless otherwise noted, $SDV_{DD}=DV_{DD}=AV_{DD}=V_P=V_{VCO}=CE=3.3V\pm 10\%$, $SD_{GND}=D_{GND}=A_{GND}=CP_{GND}=A_{GNDVCO}=0V$. $T_A = -55\text{ }^{\circ}\text{C} \sim 125\text{ }^{\circ}\text{C}$. High Voltage: 3.3 V, Low Voltage: 0 V, Timing Frequency $f_{max}=25\text{ MHz}$, $T=40\text{ ns}$.)

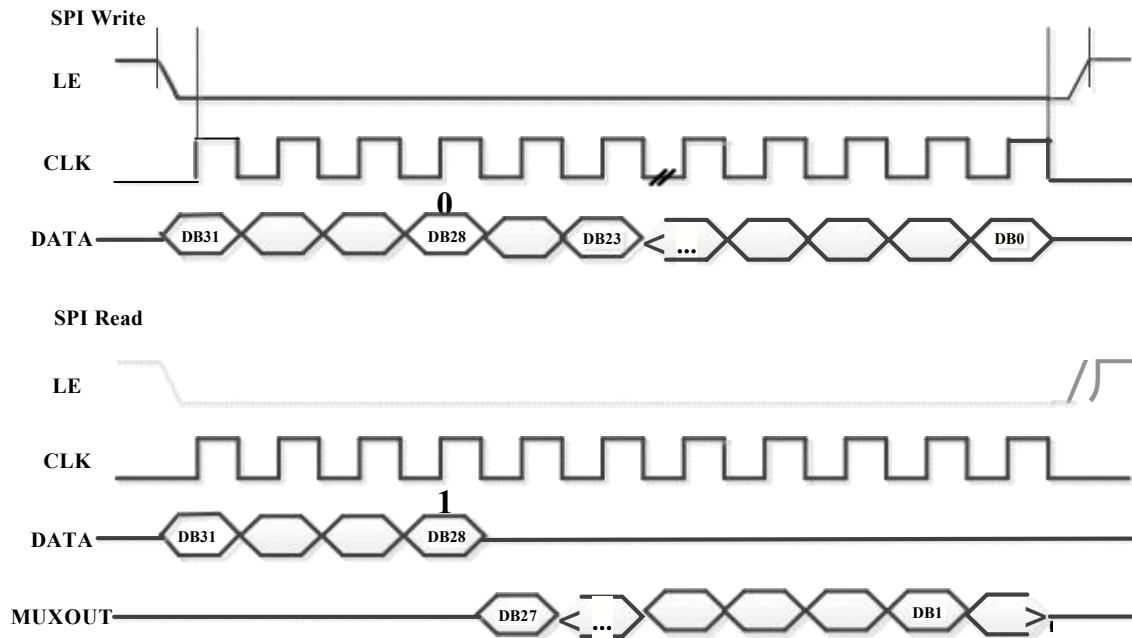


Figure 2. Timing Diagram

Note:

1. Register REG0 needs to be configured last, and all register data is refreshed.
2. The first address bit, then read and write bit, the last data bit. The address bit is high first, the data bit is high first.
3. The host sends data along the falling edge, and the slave receives data along the rising edge.
4. SPI speed up to support 25MHz.
5. It is recommended to configure 10 clock edges between registers.

REGISTER MAPS

REGO

R/W read/write
R read

REG0

CONTROL BITS																																			
R/W								SPI_CP								16_BIT INTEGER VALUE(INT)																			
ADDRESS BITS								WR	LK_SW	VCO_BIT	EN_CP																								
DB31	DB30	DB29	DB28	DB27	DB26	DB25	DB24	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0				
0	0	0	0/1	0	1	0	0/1	A7	A6	A5	A4	A3	A2	A1	A0	N15	N14	N13	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1	N0				

DB28 Read-write bit

0	Read(default)
1	Read

DB27 Lock detection mode

1	Default
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DB26 DB25 VCO Sub-band Compensation

1	0	Default
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DB24 Charge Pump Enable

0	CP turn-off
1	Cp turn-on (default)

DB23 DB22 DB21 DB20 DB19 DB18 DB17 DB16

0	0	0	0	0	0	0	1	25uA
0	0	0	0	0	0	1	0	50uA
0	0	0	0	0	1	0	0	100uA
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
0	0	0	0	0	0	0	0	
1	1	1	1	1	1	1	1	3200uA
1	1	1	1	1	1	1	1	6375uA

DB15 DB14 ■ ■ 0 **DB4 DB3 DB2 DB1 DB0**

0	0	■■■	0	0	0	0	0	NOT ALLOWED
0	0	■■■	0	0	0	0	0	■■■
0	0	■■■	0	0	0	0	0	NOT ALLOWED
0	0	■■■	1	0	1	1	0	54
0	0	■■■	1	0	1	1	1	55
0	0	■■■	1	1	0	0	0	56
■	■	■■■	■	■	■	■	■	■ ■ ■
■	■	■■■	■	■	■	■	■	■ ■ ■
1	1	■■■	1	1	1	1	1	65535

INTmin = 54 with Prescaler=4/5

REG1

R/W		CONTROL BITS																													
ADDRESS BITS		WR	FDIV_SW	CONRT3B		24_BIT FRACTIONAL VALUE(FRAC)																									
DB31	DB30	DB29	DB28	DB27	DB26	DB25	DB24	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
0	0	1	0/1	0/1	M1	M0	0/1	FR23	FR22	FR21	FR20	FR19	FR18	FR17	FR16	FR15	FR14	FR13	FR12	FR11	FR10	FR9	FR8	FR7	FR6	FR5	FR4	FR3	FR2	FR1	FR0

DB28	Read-write bit
0	Write(default)
1	Read

DB27	FDIV_SW
1	4/5 (Default)
0	8/9

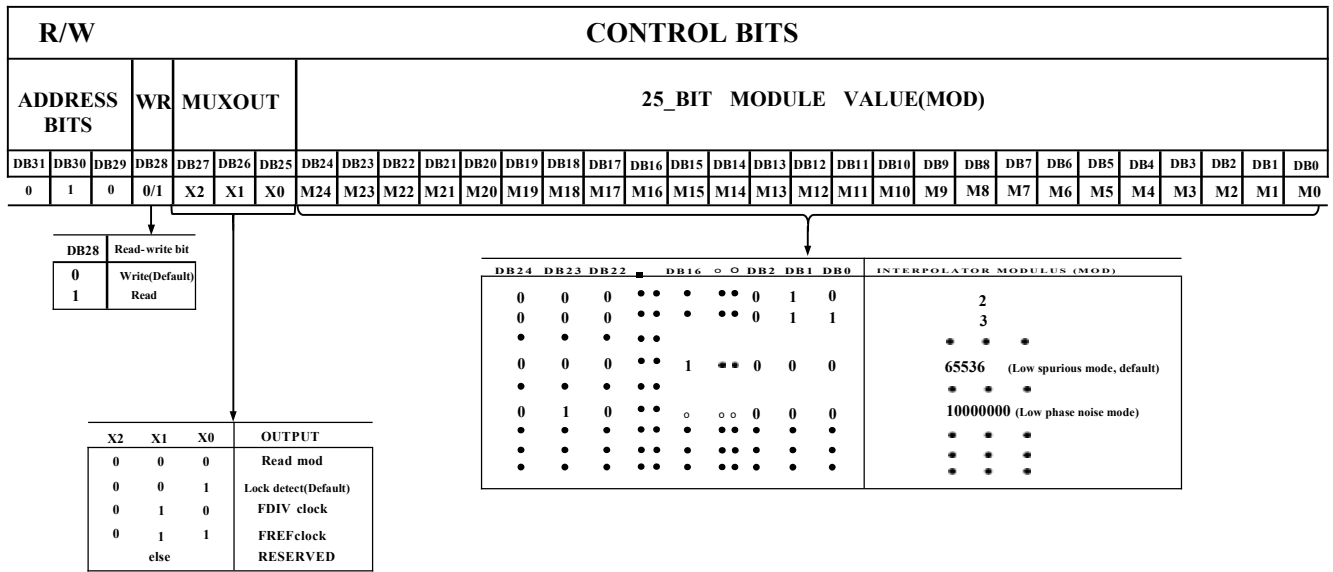
DB26	DB25	SDM mod
0	0	Traditional SDM
1	0	Scrambling SDM

DB24	CONRT3B
0	SDM turn-off
1	SDM turn-on(Default)

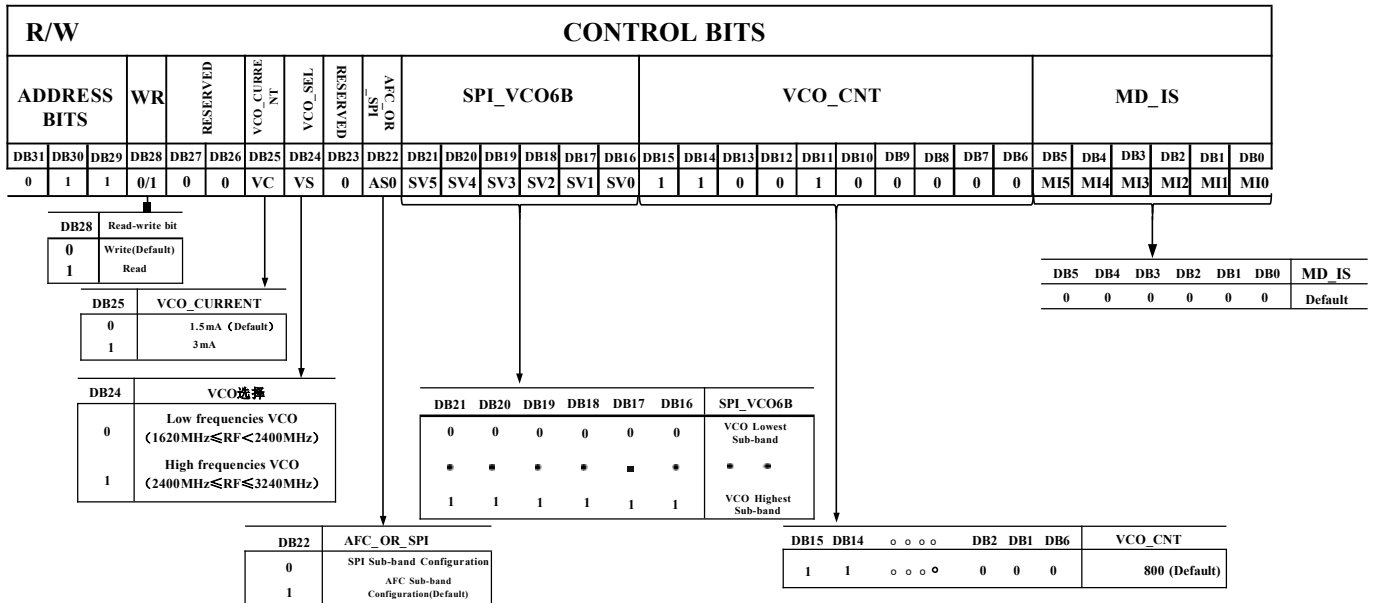
PS: FRAC24B=FRAC

DB23	DB22	0	0	0	0	DB3	DB2	DB1	DB0	FRACTIONAL VALUE (FRAC)
0	0	■	○	●	●	0	0	0	0	0
0	0	○	○	○	○	0	0	0	1	1
0	0	○	○	○	○	0	0	1	0	2
0	0	○	■			0	0	1	1	3
●	●	○	■	○	○	■	●	●	●	● ● ●
1	1	○	■	○	○	1	1	0	0	16777212
1	1	○	○	○	○	1	1	0	1	16777213
1	1	○	○	○	○	1	1	1	0	16777214
1	1	○	○			1	1	1	1	16777215

REG2



REG3



REG4

CONTROL BITS																																		
ADDRESS BITS			WR	LD_SW	AFC_VBIAS						FREF_DIV														VCO_DIVIDER									
DB31	DB30	DB29	DB28	DB27	DB26	DB25	DB24	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0			
1	0	0	0/1	0	0	1	0	0	0	0	F13	F12	F11	F10	F9	F8	F7	F6	F5	F4	F3	F2	F1	F0	SD6	SD5	SD4	SD3	SD2	SD1	SD0			

DB28	Read-write bit
0	Write(Default)
1	Read

DB27	DB26	RF output power mod
1	0	High power output
1	1	Low power output

DB20	DB19	DB18	...	DB9	DB8	DB7	Reference predivider
0	0	0	...	0	0	1	/1
0	0	0	...	0	1	0	/2
0	0	0	...	1	0	0	/4
0	0	0	...	0	0	0	/8
0	0	1	...	0	0	0	/16
...	...	...	...	...	...	...	...
1	1	1	...	1	1	1	/16383

DB6	DB5	DB4	DB3	DB2	DB1	DB0	VCO RF frequency divider
0	0	0	0	0	0	1	Shoot-through
0	0	0	0	0	1	0	/divide-by-2
0	0	0	0	1	0	0	/divide-by-4
0	0	0	1	0	0	0	/divide-by-8
0	0	1	0	0	0	0	/divide-by-16
0	1	0	0	0	0	0	/divide-by-32
1	0	0	0	0	0	0	/divide-by-64

DB25	DB24	DB23	DB22	DB21	AFC control mode
1	0	0	0	0	Default

REG5

R/W CONTROL BITS																																		
ADDRESS BITS			WR	RESERVED																				FFD_DELA_Y_D		FFD_DELA_Y_EN	LFSR_SEL		BIT_SEL					
DB31	DB30	DB29	DB28	DB27	DB26	DB25	DB24	DB23	DB22	DB21	DB20	DB19	DB18	DB17	DB16	DB15	DB14	DB13	DB12	DB11	DB10	DB9	DB8	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0			
1	0	1	0/1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	SD4	1	0	1	0			

DB28

Read-write bit

0

Write(Default)

1

Read

DB6DB5

Delayed control mode

00

Default

DB4

Optimization of Fractional Phase Noise Enablement

0

Integer(off)

1

Decimal mode(on)

DB3DB2

LFSR_SEL

10

Default

DB1DB0

BIT_SEL

10

Default

REGISTER 0

Control Bits

With Bits DB[31:29] set to 0, 0, 0, Register 0 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 0 can be written.

Charge Pump Enable Bit

The charge pump enable bit DB[24] in register 0 is used to control the charge pump current switch. Default charge pump current on DB[24] set to 1.

Charge Pump Current Set Bit

The charge pump current setting bit DB[23:16] of register 0 is used to set the charge pump current. The charge pump current should be set to the loop filter's design current of 3.2mA (see Table 2).

INT Value Set Bits

The INT setting bit DB[15:0] of register 0 is used to set the INT value, which determines the integer part of the feedback frequency dividing coefficient. Used for formulas $RF_{OUT} = [INT + (FRAC/MOD)] \times [f_{pFD}]/RF \text{ divider}$.

RF_{out} is the RF output frequency, INT is the integer part of the feedback frequency division coefficient, FRAC/MOD is the decimal part of the feedback frequency division coefficient, f_{pFD} is the phase detection frequency, RF divider is the VCO radio frequency divider 1~64 frequency division.

REGISTER 1

Control Bits

With Bits DB[31:29] set to 0, 0, 1, Register 1 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 1 can be written.

Prescaler Value Set Bit

Register 1's prescaler value set bit DB[27] is used to set 4/5 or 8/9. Since the maximum allowable RF frequency of a prescaler value based on synchronous 4/5 division is 3.24GHz, the LQ883's full-band setting defaults to 4/5 division.

SDM Mode and Switch Set Bits

SDM mode and switch set bit DB[26:24] of register 1 is used to set SDM mode and switch. The SDM switch is on by default, and can be achieved by setting the FRAC molecule value of the decimal input to 0 when the integer frequency is implemented.

FRAC Set Bit

The FRAC set bit DB[23:0] of register 1 is used to set the molecular value of the SDM decimal input. It is used in conjunction with the denominator MOD value of the SDM decimal input to achieve the fractional part of the feedback division coefficient.

REGISTER 2

Control Bits

With Bits DB[31:29] set to 0, 1, 0, Register 2 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 2 can be written.

MUXOUT Set Bits

The MUXOUT set bits DB[27:25] of register 2 is used for either the lock detect, the scaled RF, or the scaled

reference frequency to be accessed externally, with default access lock detection.

MOD Module Set Bits

The MOD module set bits DB[24:0] of register 2 is used to set the fractional modulus value of the SDM decimal input. It is used in conjunction with the molecular FRAC value of the SDM decimal input to achieve the fractional part of the feedback frequency division coefficient. When the MOD module is set DB[24:0]= 25 'd10000000, the random sequence of SDM output is more periodic, so the phase noise performance is good, and the decimal stray performance is poor; When MOD module set bit DB[24:0]= 25 'd16777216, SDM output random sequence periodic poor, so the fractional stray performance is better, the phase noise performance is poor.

REGISTER 3

Control Bits

With Bits DB[31:29] set to 0, 1, 1, Register 3 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 3 can be written.

VCO Select Set Bit

VCO Select Set bit DB[24] of Register 3 is used to select VCO outputs in different RF bands. When 1620MHz is less than or equal to the RF output frequency < 2400MHz, select the low-band VCO. When the RF output frequency is less than or equal to 3240MHz, select the high band VCO.

AFC or SPI Select Set Bit

Register 3's AFC or SPI select set bit DB[22] for Select Auto select VCO sub-band or SPI configuration Select sub-band. When DB[22]= 1'b1, the VCO sub-band corresponding to the RF output frequency is automatically selected through the AFC automatic frequency calibration function. When DB[22]= 1'b0, the SPI is used to manually configure the VCO sub-band corresponding to the RF output frequency. This function must be used together with the VCO sub-band 6bit control bit.

VCO Sub-band 6bit Control Bits

The VCO sub-band 6bit control bits DB[21:16] of register 3 is used for the selection of different sub-bands of VCO 64, which needs to be used in conjunction with the AFC or SPI select setting bit. For example, if the RF output frequency is 2GHz and the corresponding low-band VCO sub-band control bit in the table is 6 'b101000, DB[23:22]= 2' b00 and DB[21:16]= 6 'b101000 need to be configured in register 3. When DB[23:22]= 2 'b11, the VCO sub-band 6bit control bits are invalid.

REGISTER 4

Control Bits

With Bits DB[31:29] set to 1, 0, 0, Register 4 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 4 can be written.

RF Output Power Mode Set Bits

The RF output Power mode set bits DB[27:26] in register 4 is used to select different RF power output modes. When DB[27:26]=11, it is low power RF output mode; When DB[27:26]=10, high power RF output mode.

Reference Prescaler Set Bits

The reference prescaler set bit DB[20:7] of register 4 is used to prescale the input reference clock frequency (REFIN) to meet the phase detector input frequency range.

VCO RF Prescaler Set Bits

The VCO RF prescaler set bits DB[6:0] of register 4 is used to prescale the RF output frequency by 1 to 64 to achieve a wider band range.

REGISTER 5

Control Bits

With Bits DB[31:29] set to 1, 0, 1, Register 5 is programmed.

Read/Write Bit

With Bit DB[28] set to 0, Register 5 can be written.

Phase Detection Delay Enable Set Bit

The phase detection delay enable set bit DB[4] of register 5 is used to optimize the fractional mode phase noise, and DB[4] is set to 0 when the frequency division ratio is in integer mode; DB[4] is set to 1 when the frequency division is in decimal mode.

The sequence of configuration for REGISTER 0~5

When the power pin application place 4.7uF, 0.1uF, 0.01uF capacitors, maximize the interference on the power line filter and power on normal, LQ883in accordance with the following register order to complete the configuration:

Register 5→ Register 4→ Register 3→ Register 2→

Register 1→ register 0

Frequency Point Configuration Instructions

Here is a Application example of how to configure the LQ883:

$$RF_{OUT} = [INT + (FRAC/MOD)] \times [f_{PFD}] / RF \text{ divider}$$

RF_{out} is the RF output frequency, INT is the integer part of the feedback frequency division coefficient, $FRAC$ is the fractional frequency division coefficient, and MOD is the modulus, f_{PFD} is the phase detection frequency, $RF \text{ divider}$ is the VCO radio frequency divider 1~64 frequency division.

$f_{PFD} = REFIN \times [1/R]$ $REFIN$ is the reference frequency input, R is the reference prescaler coefficient.

Assume that the existing RF system needs to output frequency of 1620.2MHz (RF_{OUT}), the reference crystal ($REFIN$) frequency of 20MHz, and the frequency channel resolution (f_{RESOUT}) of 200KHz. From this we have:

LQ883's VCO output frequency 1620 ~ 3240MHz, so $RF \text{ divider} = 1$; The output requirement is 200KHz channel resolution(f_{RESOUT}), so the VCO output channel resolution is also 200KHz (f_{RES}).

$$MOD = 16777216$$

$$f_{PFD} = 20\text{MHz} \times [(1)/(1)] = 20\text{MHz}$$

$$1620.2 = 20 \times [INT + (\frac{FRAC}{16777216})]$$

$$\text{Available: } INT = 81$$

$$FRAC = 167772$$

In the above calculation, the MOD value is 16777216.

In the calculation of f_{PFD} , the reference divider coefficient R is 1. The value of f_{PFD} here can be set flexibly, but it is less than the maximum frequency of the PFD.